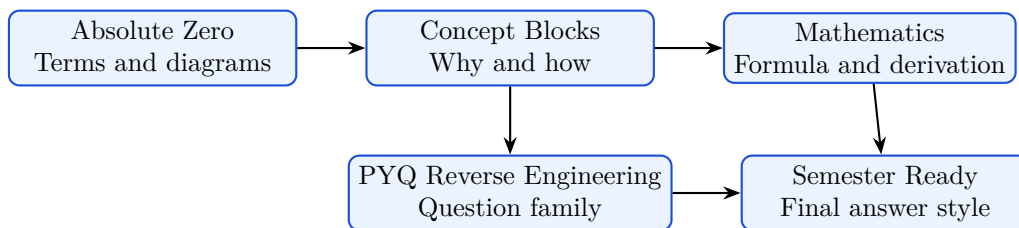


Power Supply Systems

Part-II Complete Roadmap

From Absolute Zero to Semester-Question Ready

**SN Ma'am Style: Simple English, Clean Diagrams,
Stepwise Derivations, PYQ Reverse Engineering**



Prepared for: Second Year Electrical Engineering
Power Supply Systems - Part II

Academic Article / Printed Book Style Roadmap

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1. What exactly is Part-II?

Part-II syllabus in one line

Part-II is the practical delivery side of power supply. It starts after power is generated and mainly studies how electrical power is distributed, protected, wired, earthed and economically designed.

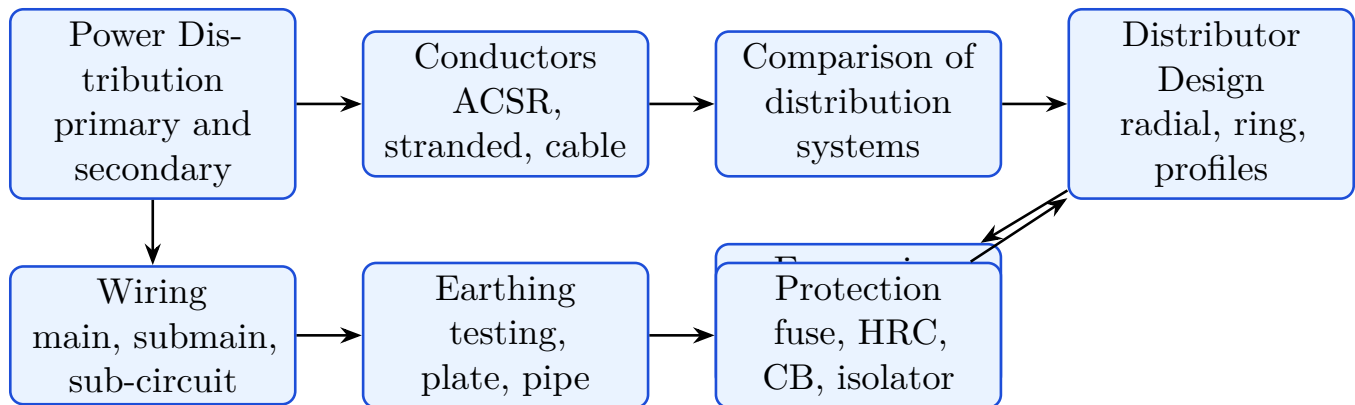


Figure 1: Complete Part-II syllabus map.

Memory hook

Remember Part-II by this chain:

Distribute → Design → Protect → Wire → Earth

If a question comes from Part-II, first ask: Is it about distribution, design, protection, wiring, or earthing?

2. Best sequence to learn the whole Part-II

Phase	Topic block	What you must become able to do	PYQ form
0	Language of power dis-tribution	Know source, feeder, distributor, service mains, voltage drop, loss, load point.	Short definitions
1	Distribution system ba-sics	Draw primary/secondary distribution and compare radial/ring/interconnector.	Theory 4–6 marks
2	Conductors and con-ductor comparison	Compare copper volume for DC/AC sys-tems; understand overhead vs under-ground basis.	Derivation/numerical
3	Kelvin's law and eco-nomics	Prove economic conductor size and solve economic area/current-density numeri-cals.	4+8 marks
4	DC distributor design	Solve fed-one-end, both-end-fed, uniform load, concentrated load, ring main.	Main numerical
5	Substation and busbar	Explain substation equipment, isolator, CB, instrument transformer, busbar.	Group B theory
6	Fuse and protection	Explain fuse, HRC fuse, pre-arcing, re-lays, circuit breaker action.	Short + diagram

7	Conductors and earthing	Stranded conductor, ACSR, plate/pipe earthing, earth resistance factors.	Short + diagram
8	Wiring and service mains	LT/MT/HT service main, sub-circuit wiring, general stipulations.	5–7 marks
9	Final reverse engineering	Identify question family and write answer in SN ma'am format.	Full paper ready

SN ma'am answer format to follow everywhere

Definition → Diagram → Assumption → Formula → Steps → Final answer

Never start a numerical directly from formula. First draw the physical system and mark current directions.

3. Phase 0: Absolute-zero vocabulary

3.1 Feeder, distributor and service mains

Definitions

- **Feeder**: A line which carries power from substation to distribution area. Normally no tapping is taken from it. It is designed mainly by current-carrying capacity.
- **Distributor**: A line from which many tappings are taken for consumers. It is designed mainly by permissible voltage drop.
- **Service mains**: Conductors which connect consumer terminals to the distributor.

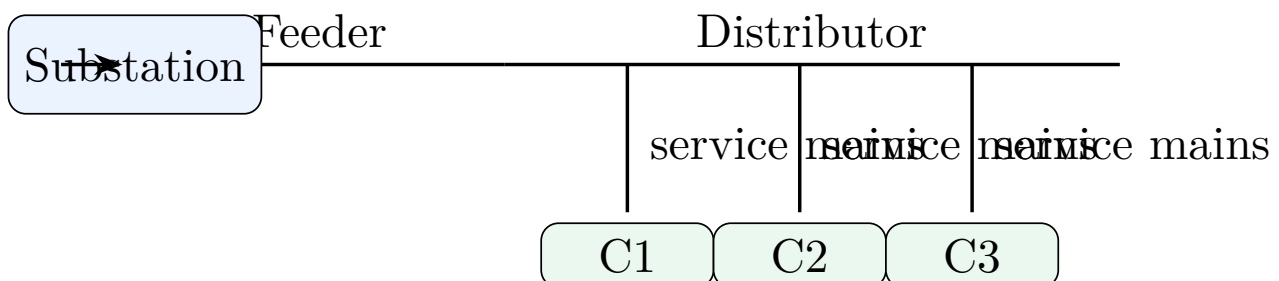


Figure 2: Feeder, distributor and service mains.

Memory hook

Feeder = current capacity, Distributor = voltage drop, Service mains = consumer connection

4. Phase 1: Distribution systems

4.1 Primary and secondary distribution

Basic idea

After transmission, voltage is stepped down in substations and supplied through distribution lines. Primary distribution is usually at higher distribution voltage such as 11 kV. Secondary distribution supplies consumers at utilisation voltage such as 400 V three-phase or 230 V single-phase.

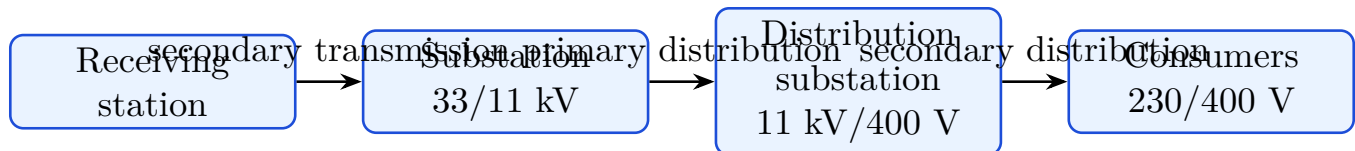


Figure 3: Primary and secondary distribution chain.

4.2 Radial, ring main and interconnector

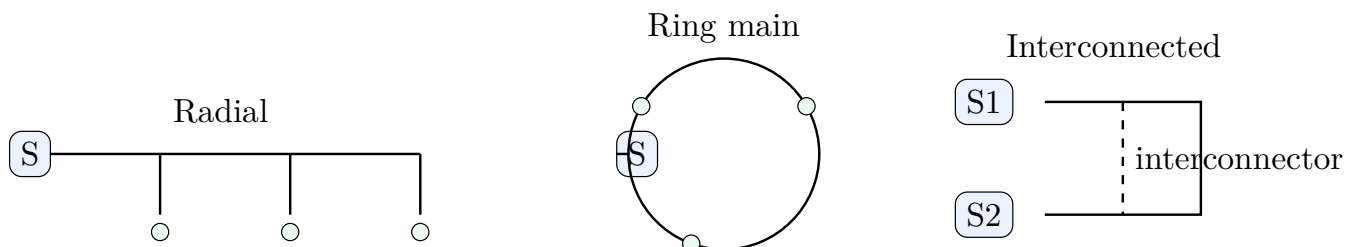


Figure 4: Radial, ring main and interconnected distribution systems.

System	Simple meaning	Advantage	Weak point
Radial	Supply comes from one end only.	Simple and cheap.	Poor reliability; far end voltage is low.
Ring main	Distributor forms a closed loop.	Better voltage and continuity.	Cost and operation are higher.
Interconnected	Two or more sources are tied.	Highest reliability and load sharing.	Protection and control are more complex.

Most predictable questions

- Explain why loop arrangement is better than radial arrangement.
- Differentiate between radial system, ring main system and ring main with interconnector.
- Explain the basis of design of feeders and distributors.

5. Phase 2: Conductors and comparison of distribution systems

5.1 Overhead and underground comparison basis

Most important assumption

For overhead lines, maximum voltage between conductor and earth is used as basis. For underground cables, maximum voltage between conductors is used as basis because insulation stress occurs between conductors.

5.2 Master method for conductor-material comparison

Never memorise blindly

Every conductor-comparison derivation follows one chain:

$$P = VI \cos \phi \quad \text{or} \quad P = \sqrt{3}V_L I_L \cos \phi$$

$$P_{loss} = nI^2 R, \quad R = \frac{\rho l}{a}, \quad \text{Volume} = nal$$

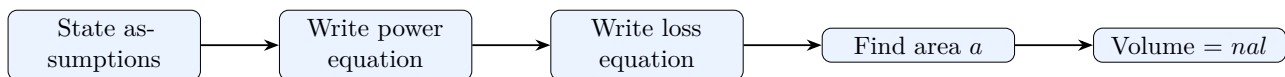


Figure 5: Reverse-engineering method for conductor-comparison questions.

5.3 Diagrams to practise

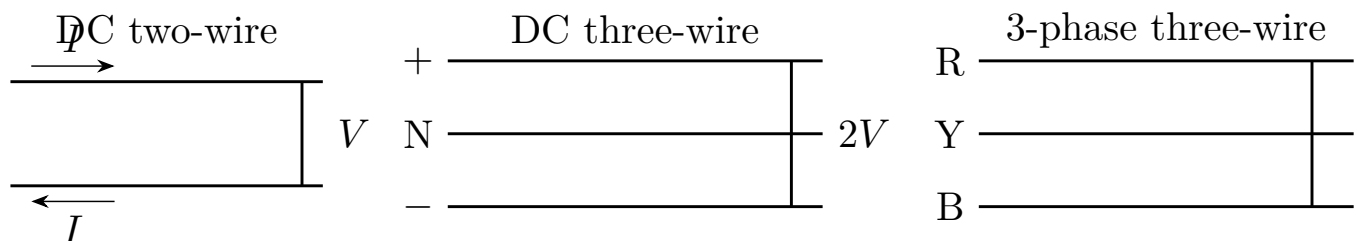


Figure 6: Basic conductor systems used in comparison questions.

Semester-like example family

Question form: Compare the volume of conductor material required for DC three-wire system with AC two-phase three-wire system in underground cables. State assumptions.

What to do: Draw both systems, state assumptions, use same P , l , P_{loss} and voltage basis, then compare nal .

6. Phase 3: Kelvin's law and feeder economics

Definition

Kelvin's law states that the most economical size of conductor is obtained when the variable annual charge on capital cost is equal to the annual cost of energy wasted in the conductor.

6.1 Derivation

Let a be the conductor cross-sectional area. The annual capital charge can be written as

$$C_1 = P_1 + P_2 a$$

The annual cost of energy wasted decreases with area because resistance is inversely proportional to area:

$$R = \frac{\rho l}{a}$$

So

$$C_2 = \frac{P_3}{a}$$

Total annual cost:

$$C = P_1 + P_2 a + \frac{P_3}{a}$$

For minimum cost,

$$\frac{dC}{da} = 0$$

$$P_2 - \frac{P_3}{a^2} = 0$$

$$P_2 a = \frac{P_3}{a}$$

Therefore,

Variable annual capital charge = Annual cost of energy loss

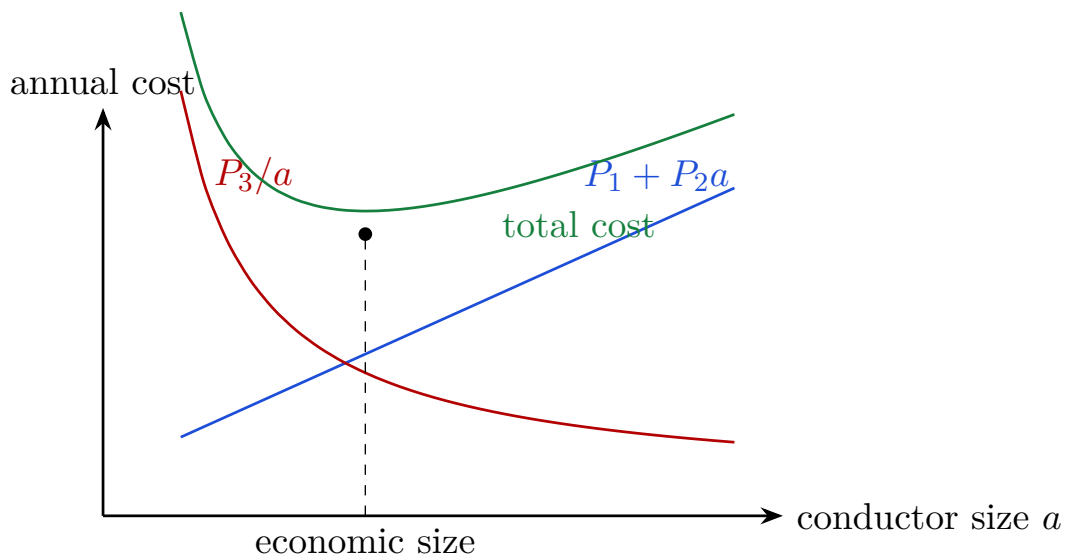


Figure 7: Kelvin's law cost curve.

6.2 Numerical algorithm

How to solve Kelvin's law numerical

1. Write line data, voltage, power, time and cost of energy.
2. Find current using $I = P/(\sqrt{3}V \cos \phi)$ for three-phase.
3. Write $R = \rho l/a$ or use resistance per km for 1 cm^2 conductor.
4. Calculate annual energy loss in kWh.
5. Convert energy loss to annual money loss.

6. Calculate variable annual charge on conductor cost.
7. Equate by Kelvin's law and find a .

Limitations

Kelvin's law is not strictly followed because actual load curve is difficult to estimate, mechanical strength and corona are not included, voltage drop may decide conductor size, and interest/depreciation are not exactly known.

7. Phase 4: DC distributor design

7.1 Distributor fed at one end

Concept

When a distributor is fed at one end, current is maximum near the feeding point and decreases after each load tapping. The minimum voltage occurs at the farthest point.

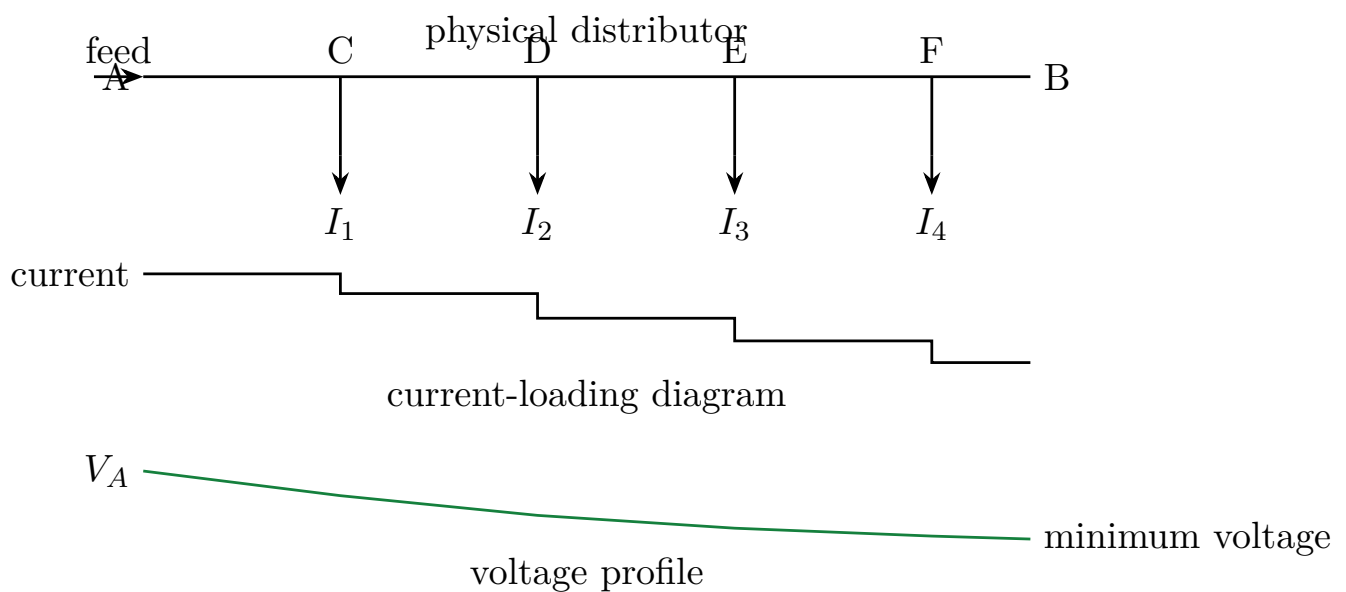


Figure 8: Distributor fed at one end with current-loading and voltage profile.

7.2 Master formula for concentrated loads

For loads I_1, I_2, I_3, I_4 at sections with resistances r_1, r_2, r_3, r_4 :

$$\Delta V = (I_1 + I_2 + I_3 + I_4)r_1 + (I_2 + I_3 + I_4)r_2 + (I_3 + I_4)r_3 + I_4r_4$$

7.3 Distributor fed at both ends

Concept

When a distributor is fed from both ends, current comes from both sides. The voltage first decreases, reaches a minimum point, then increases towards the other end.

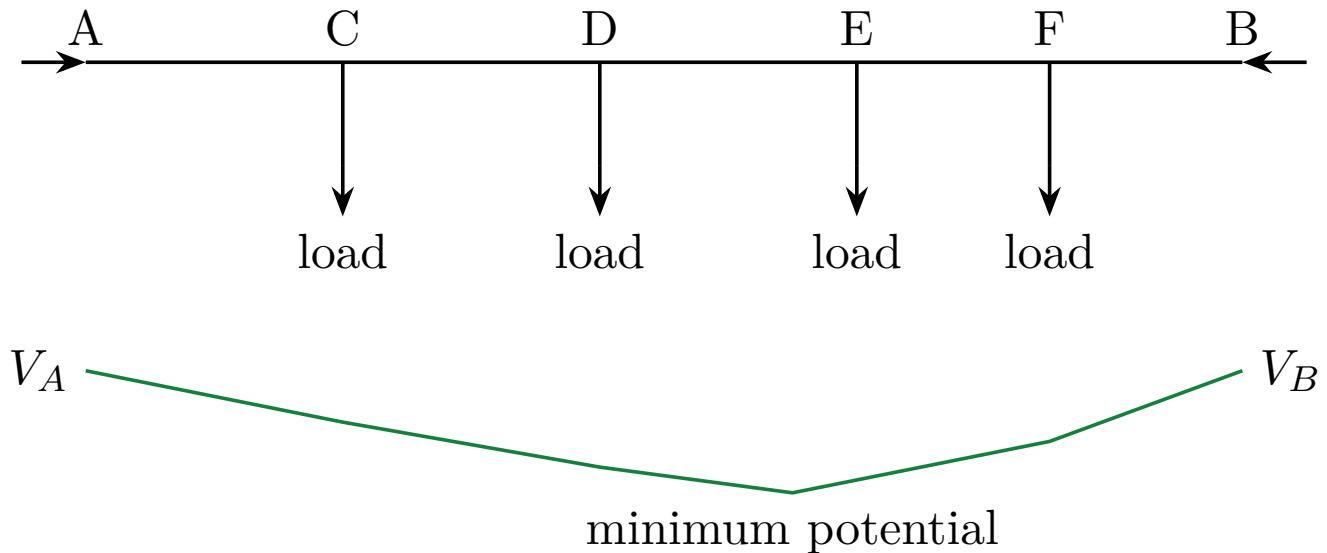


Figure 9: Voltage profile of distributor fed at both ends.

Both-end-fed concentrated load method

1. Assume current I_A from end A.
2. Write currents in all sections by subtracting tapped loads.
3. Apply $V_A - V_B = \sum IR$.
4. If $V_A = V_B$, then $\sum IR = 0$.
5. Find I_A and section currents.
6. The point where current changes sign is the point of minimum potential.

7.4 Uniformly loaded distributor fed at both ends

If load is i A/m, length is l , resistance per metre is r , and both end voltages are equal, maximum drop occurs at the middle:

$$\Delta V_{max} = \frac{ir l^2}{8} = \frac{IR}{8}$$

where $I = il$ and $R = rl$.

7.5 Ring main distributor**Concept**

A ring distributor is a closed distributor. For calculation, cut it mentally at the feeding point. It becomes equivalent to a distributor fed at both ends with equal voltage.

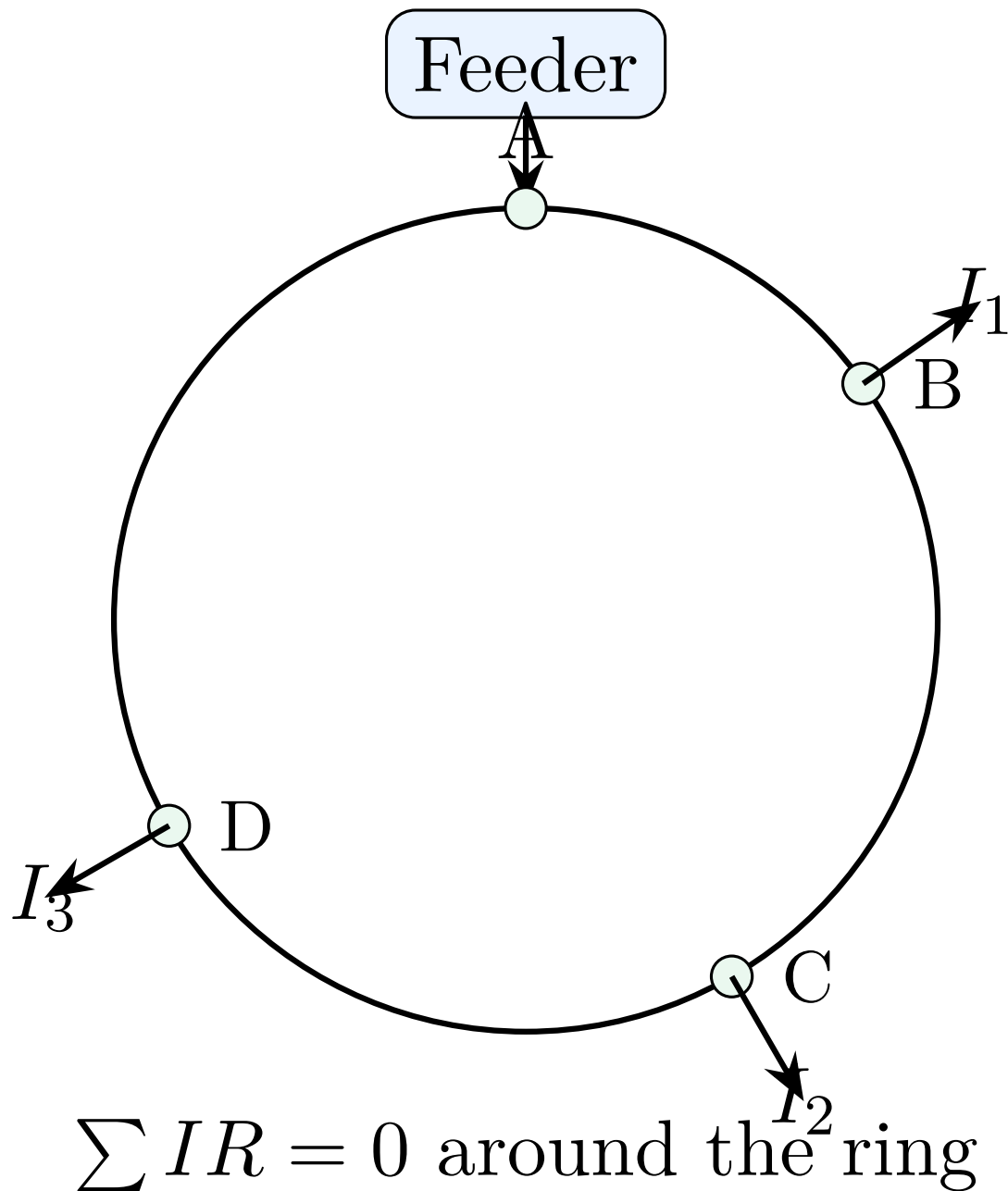


Figure 10: Ring main distributor - KVL based calculation.

8. Phase 5: Substation and busbar

8.1 Substation

Definition

A substation is an assembly of electrical equipment used to change some characteristic of electric supply such as voltage level, switching, protection, power factor or form of supply.

8.2 Major equipment

Equipment	Function
Power transformer	Steps voltage up or down.

Busbar	Common conductor to which incoming and outgoing circuits are connected.
Circuit breaker	Opens the circuit automatically under fault condition.
Isolator	Provides visible isolation after circuit breaker opens.
Lightning arrester	Protects equipment from surge voltage.
CT and PT	Step down current and voltage for metering and protection.
Relay	Detects abnormal condition and sends trip signal to circuit breaker.

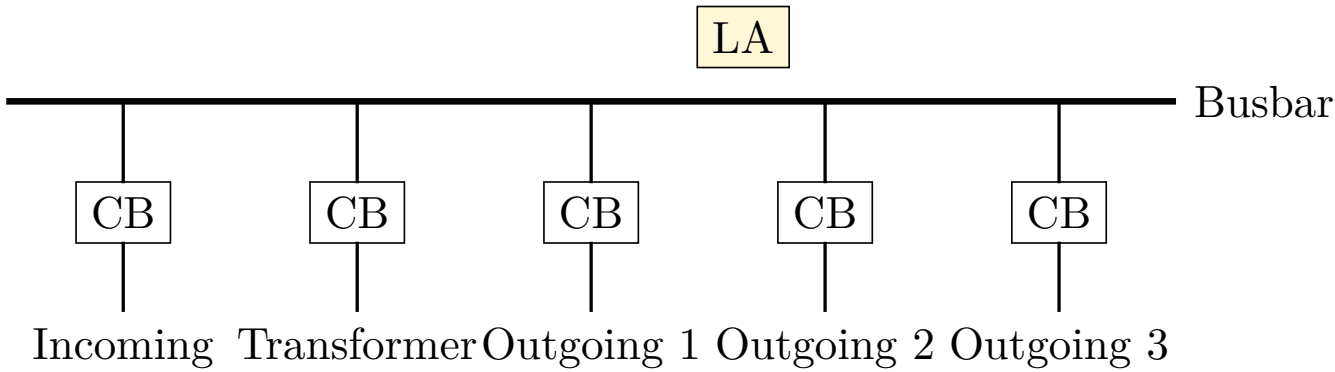


Figure 11: Single busbar arrangement.

8.3 Double busbar

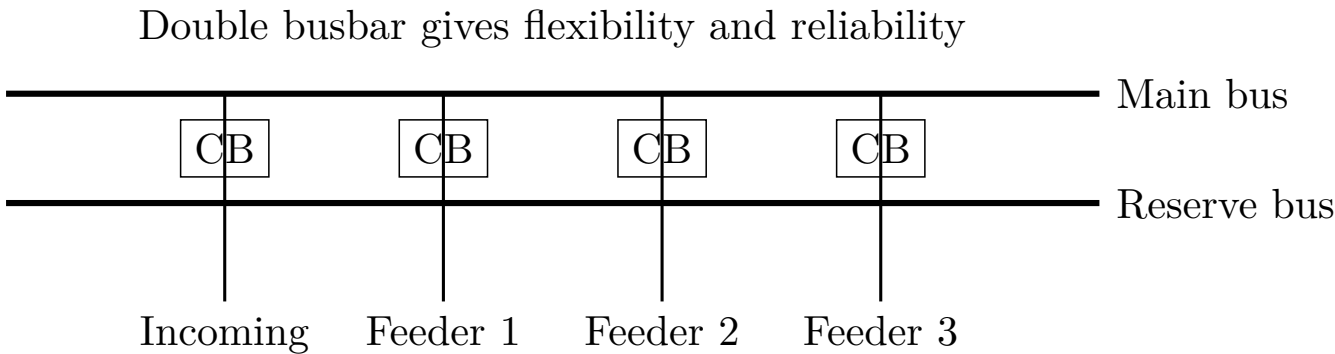


Figure 12: Simplified double busbar arrangement.

9. Phase 6: Fuse, HRC fuse and relay

9.1 Fuse

Definition

A fuse is a short piece of metal connected in series with the circuit. It melts when excessive current flows and breaks the circuit.

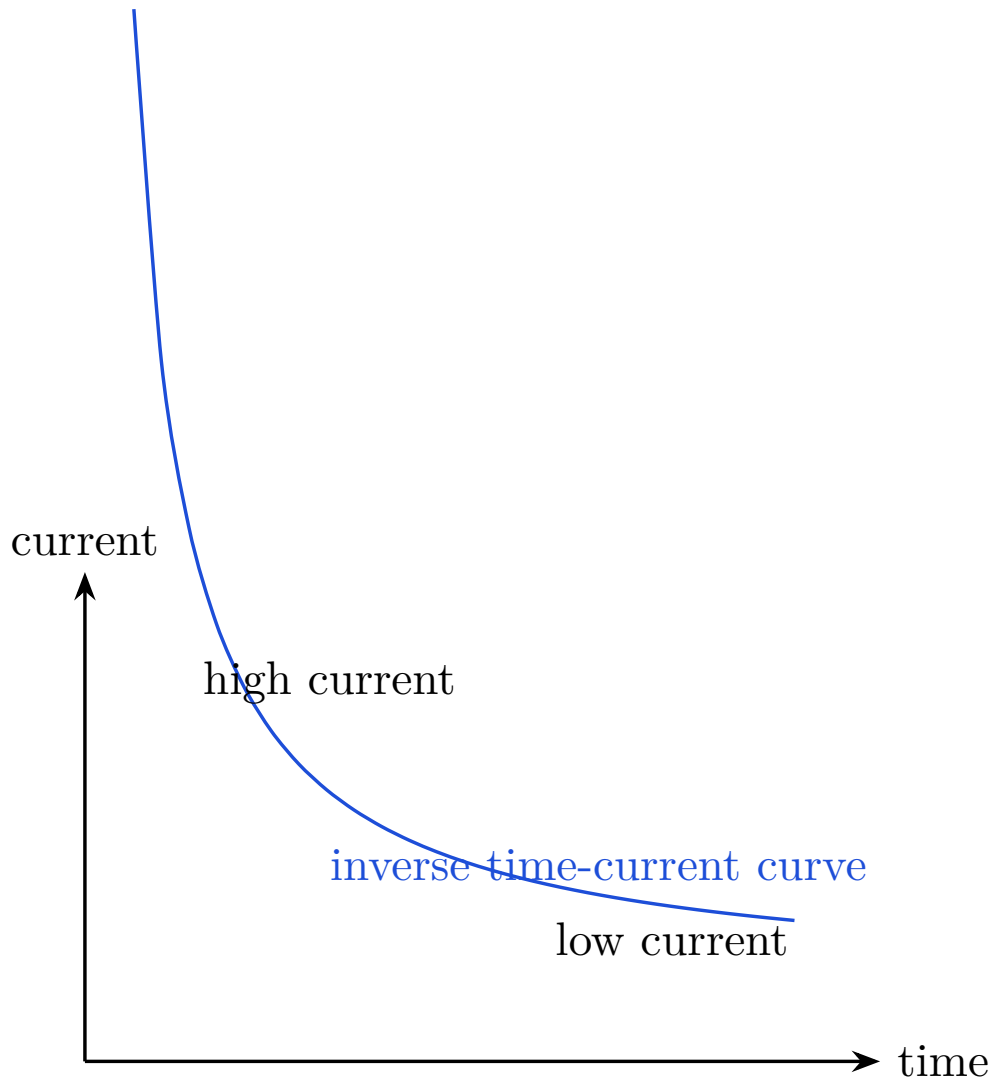


Figure 13: Fuse inverse time-current characteristic.

9.2 HRC fuse

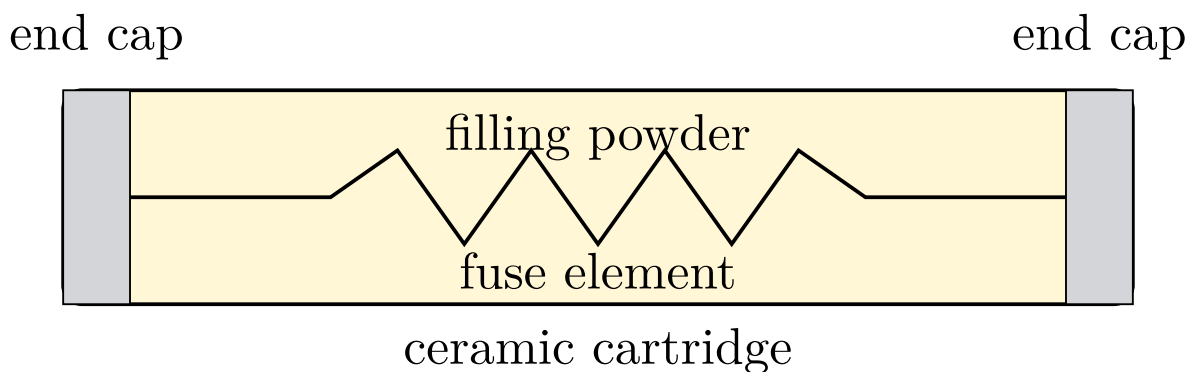


Figure 14: HRC fuse construction.

9.3 Protective relay operation

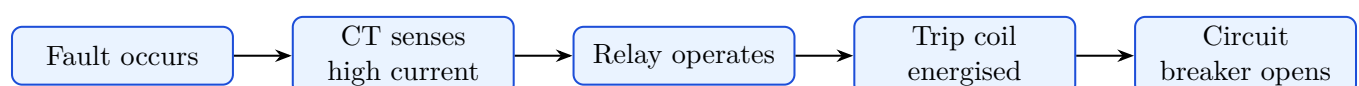


Figure 15: Protective relay operation flowchart.

10. Phase 7: Conductors and earthing

10.1 Stranded conductor

Definition

A stranded conductor is made by twisting many thin wires called strands. It is flexible and easy to coil and transport.

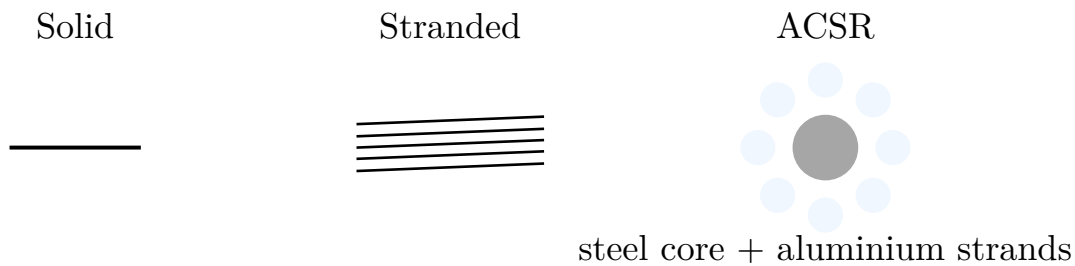


Figure 16: Solid, stranded and ACSR conductor idea.

10.2 Earthing

Definition

Earthing means connecting the metallic body of electrical equipment to the general mass of earth through a low resistance path.

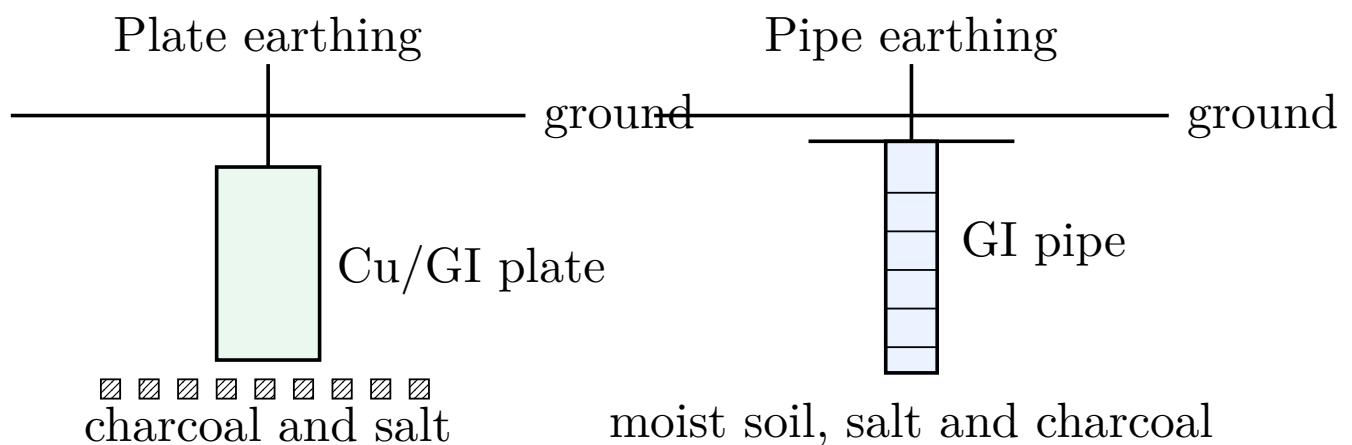


Figure 17: Plate and pipe earthing.

Most predictable questions

- What is earthing? Mention factors affecting earth resistance.
- Explain plate earthing with diagram.
- Explain pipe and plate earthing.
- Mention advantages of neutral grounding.

11. Phase 8: Wiring and service main connection

11.1 Low tension service main

Basic idea

The service main connects the supply company's line to the consumer's installation. In low tension wiring, safety, voltage drop, protection and proper distribution of load are most important.

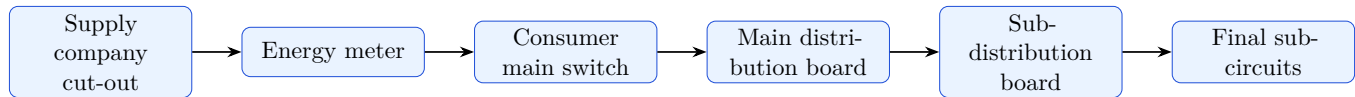


Figure 18: Service main connection flow.

11.2 General stipulations

- Power and heating circuits should be separate from lighting and fan circuits.
- Voltage drop should not exceed the allowed limit.
- Fuse should be placed in live conductor, not in neutral.
- Neutral wire should not be broken by switch or fuse.
- Each sub-circuit should have limited number of points and limited load.
- All metal boards and frames should be properly earthed.

11.3 Sub-circuit wiring systems

System	Meaning	Main point
Tree system	Main cable goes around the building and loads are tapped.	Flexible but more cable is required.
Looping-in system	Live wire is looped inside switch-board.	Cheap and commonly used.
Junction-box system	Connections are made in junction boxes.	Costly but fault isolation is better.

12. Phase 9: PYQ reverse-engineering table

If the question says...	It belongs to...	How to attack
"Compare volume of conductor material"	Phase 2	State assumptions, draw systems, use P , P_{loss} , V , volume = nal .
"State/prove Kelvin's law"	Phase 3	Write cost equation, differentiate, draw cost curve, state limitation.
"DC distributor with loads"	Phase 4	Draw line, assume currents, write section currents, calculate drops.
"Fed at both ends"	Phase 4	Use $V_A - V_B = \sum IR$, find current reversal point.
"Ring main"	Phase 4	Cut ring mentally, apply $\sum IR = 0$, find voltage at load points.

"HRC fuse"	Phase 6	Define fuse, draw cartridge, explain arc quenching and advantages.
"Relay operation"	Phase 6	Fault $\rightarrow$ CT $\rightarrow$ relay $\rightarrow$ trip coil $\rightarrow$ CB.
"Busbar arrangement"	Phase 5	Draw busbar, label CB/isolator/LA, give merits/demerits.
"Earthing"	Phase 7	Define, purpose, diagram, factors affecting earth resistance.
"Service main connection"	Phase 8	Draw service flow and write stipulations.

13. Final semester-ready revision plan

10-day compact plan

Day	Block	Output expected
1	Feeder, distributor, service mains, primary/secondary distribution	Draw supply and distribution diagrams from memory.
2	Radial, ring, interconnector	Write comparison answers.
3	Conductor comparison basics	Derive 2-3 common systems.
4	Kelvin's law	Prove law, draw curve, solve one numerical.
5	Distributor fed one end	Solve concentrated and uniform load examples.
6	Both-end-fed and ring distributor	Solve KVL-based examples.
7	Fuse, HRC, relay	Draw and explain all protection diagrams.
8	Substation and busbar	Draw single/double busbar and equipment table.
9	Earthing, stranded conductor, ACSR, service mains	Prepare Group-B short answers.
10	Mock paper	Pick one PYQ paper and solve Part-II fully.

One-page final memory map

FDS = Feeder-current, Distributor-voltage, Service-consumer

Kelvin = capital charge = energy loss cost

Distributor numerical = Draw $\rightarrow$ section currents $\rightarrow \sum IR \rightarrow$ voltages

Protection = Fault $\rightarrow$ relay $\rightarrow$ circuit breaker

Earthing = low resistance path to earth

Closing instruction

For every semester answer, write like this:

Start simple. Draw neatly. Explain physically. Derive slowly. Solve step by step. Box the answer.